Variable Selection for Wasserstein Robust Hypothesis Testing

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Abstract

Variable selection for Wasserstein robust testing admits an exact finite formulation and a maximum-affine dual representation. Dual couplings generate global cuts. The analysis includes certificates for inexact solves and data-dependent bounds on the evaluated selections.

1 Problem formulation

Let $[D] = \{1, \dots, D\}$ and let $0 \le d \le D$ be an integer. For a measurable score $f : \mathbb{R}^D \to \mathbb{R}$, define

$$R(f) = \sup_{\substack{P_0 \in \mathcal{U}_0 \\ P_1 \in \mathcal{U}_1}} \left\{ \mathbb{E}_{P_0} \ell(-f(X)) + \mathbb{E}_{P_1} \ell(f(X)) \right\}. \tag{1}$$

The loss ℓ is finite, nonnegative, convex and nondecreasing, with $\ell(0) > 0$. Write $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let $\mathrm{Proj}_S : \mathbb{R}^D \to \mathbb{R}^{|S|}$ be the coordinate projection and define

$$\mathcal{G}_S = \{ g : \mathbb{R}^{|S|} \to \mathbb{R} : \ g \text{ is Borel measurable} \}.$$

The variable-selection problem is

$$\boxed{V(S) = \inf_{g \in \mathcal{G}_S} R(g \circ \mathrm{Proj}_S), \qquad \mathrm{OPT} = \min_{|S| \le d} V(S).} \tag{2}$$

The N observations are $x_1, \dots, x_N \in \mathbb{R}^D$. Nonempty index sets I_0, I_1 partition $[N]$ by class, with within-class masses $q_i > 0$:

$$\sum_{i \in I_k} q_i = 1, \qquad \widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \mathcal{U}_k = \{ P : W_1(P, \widehat{P}_k) \le \theta_k \}, \quad k = 0, 1.$$

Here δ_x is unit mass at x, $\theta_k > 0$, and W_1 uses ℓ_1 cost on distributions with finite first moments. Unweighted samples have $q_i = 1/|I_k|$ for $i \in I_k$.

2 Dual reformulation

Fix S and write

$$x_i^S = \mathrm{Proj}_S x_i, \qquad \Delta_{ir,j} = |x_{ij} - x_{rj}|, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1 = \sum_{j \in S} \Delta_{ir,j}.$$

Here $i, r \in [N]$ and $j \in [D]$. A coupling $\pi_{ir} \ge 0$ transports mass from source i to target r. Define

$$\mathcal{T}_k(S) = \left\{ \pi \ge 0 : \sum_r \pi_{ir} = q_i, \quad \sum_{i \in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

With target-score variables $u = (u_1, \dots, u_N)$, the finite transport representation is

$$V(S) = \inf_{u \in \mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi \in \mathcal{T}_k(S)} \sum_{i \in I_k, r} \pi_{ir} \ell_k(u_r). \tag{3}$$

Dualizing each transport LP introduces source-loss bounds $a_i \in \mathbb{R}$ and transport prices $\lambda_k \ge 0$. With $J(a, \lambda) = \sum_i q_i a_i + \sum_k \theta_k \lambda_k$, this gives

$$\begin{aligned} V(S) = \min_{a, u \in \mathbb{R}^N, \ \lambda \ge 0} \quad & J(a, \lambda) \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k, \ r \in [N], \ k = 0, 1. \end{aligned} \tag{4}$$

Appendix A proves exactness and recovers an optimal test.

For the binary indicator $z = \mathbf{1}_S$, replace the switched prices $\lambda_k z_j$ by coordinate prices w_{kj}. Since $a_i \geq 0$,

$$\sum_k \theta_k w_k = z \sum_k \theta_k \lambda_k \leq zJ(a,\lambda) \quad \text{when } w_k = \lambda_k z.$$

For fixed $z \in [0,1]^D$, define

$$\begin{aligned} F(z) = \min_{a,u,\lambda,w} \quad & J(a,\lambda) \\ \text{subject to} \quad & \ell_k(u_r) \leq a_i + \sum_j \Delta_{ir,j} w_{kj}, \quad i \in I_k,\ r \in [N],\ k = 0,1, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0,1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq zJ(a,\lambda), \\ & \lambda \geq 0, \qquad a,u \in \mathbb{R}^N. \end{aligned} \tag{5}$$

Here $w_k = (w_{k1}, \ldots, w_{kD})^\top$ and $\mathbf{1} = (1, \ldots, 1)^\top$; vector inequalities are coordinatewise. For binary z, under the other constraints,

$$\theta_0 w_0 + \theta_1 w_1 \leq zJ \quad \Leftrightarrow \quad w_{kj} = 0 \quad \text{for } z_j = 0,\ k = 0,1.$$

When $z_j = 1$, that particular inequality is redundant and may be omitted:

$$\theta_0 w_{0j} + \theta_1 w_{1j} \leq \sum_k \theta_k \lambda_k \leq J.$$

Deleting the entire family also removes the zero-coordinate restrictions, giving $V([D])$ for every z. Retaining it gives $F(\mathbf{1}_S) = V(S)$; Proposition B.1(i) proves equality of the projected feasible sets and preservation of the objective.

A dual of (5), with a scaled transport coupling $\Pi \geq 0$, a scalar $\alpha \geq 0$, and coordinate multipliers $\mu \geq 0$, is

$$\begin{aligned} F(z) = \max_{\Pi,\alpha,\mu} \quad & H(\Pi) \\ \text{subject to} \quad & (\Pi,\alpha,\mu) \in \mathcal{D}, \qquad \alpha + \mu^\top z = 1. \end{aligned} \tag{6}$$

Here

$$H(\Pi) = \sum_{r=1}^{N} \inf_{s \in \mathbb{R}} \sum_{k=0}^{1} \sum_{i \in I_k} \Pi_{ir} \ell_k(s),$$

and, with $(v)_+ = \max\{v, 0\}$,

$$\mathcal{D} = \left\{ (\Pi,\alpha,\mu) \geq 0 : \begin{array}{l} \displaystyle\sum_r \Pi_{ir} = \alpha q_i, \quad i \in [N], \\ \displaystyle\sum_j \left(\sum_{i \in I_k, r} \Pi_{ir} \Delta_{ir,j} - \theta_k \mu_j \right)_+ \leq \alpha \theta_k, \quad k = 0,1 \end{array} \right\}. \tag{7}$$

The coupling has source masses αq_i; H minimizes its weighted loss over one score per target. The Lagrangian derivation is in Proposition B.1(ii).

For a normalized tuple $\alpha + \mu^\top z = 1$, scaling all three dual variables by $\tau \geq 0$ preserves $\mathcal{D}$ and gives

$$H(\tau\Pi) = \tau H(\Pi), \qquad \max_{\tau \geq 0} \{2\sqrt{\tau H(\Pi)} - \tau\} = H(\Pi), \quad \tau^* = H(\Pi).$$

Optimizing this scale on every ray removes the normalization:

$$\boxed{F(z) = \max_{(\Pi,\alpha,\mu) \in \mathcal{D}} \{2\sqrt{H(\Pi)} - \alpha - \mu^\top z\}.} \tag{8}$$

Proposition B.1(iii) proves this equivalence and attainment. A normalized optimizer $(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$ gives the affine optimizer $F(z)(\bar{\Pi}, \bar{\alpha}, \bar{\mu})$; the scaling must also be applied to the cut slope.

3 Cutting-plane algorithm

Let $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. At z^t, evaluate $F(z^t)$ by (8) and take an optimizer (Π^t, α^t, μ^t) with the bounds $\mu_j^t \le M_j$ from Proposition C.1(ii). Its subgradient $-\mu^t$ gives

$$F(z) \ge 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z = F(z^t) - (\mu^t)^\top (z - z^t). \tag{9}$$

The next selection and lower bound solve

$$\begin{aligned} (z^{t+1}, \eta^{t+1}) \in \underset{z \in \mathcal{Z}_d,\ \eta \ge 0}{\arg\min} \quad & \eta \\ \text{subject to} \quad & \eta \ge F(z^s) - (\mu^s)^\top (z - z^s), \qquad s = 1, \dots, t. \end{aligned} \tag{10}$$

For exact solves, writing $U_t = \min_{s \le t} F(z^s)$ and T for the last iteration, Appendix C proves

$$\begin{aligned} & \eta^{t+1} \le \mathrm{OPT} \le U_t, \\ & 0 \le R(f^*) - \mathrm{OPT} \le U_T - \eta^{T+1} \le \varepsilon, \qquad T \le \sum_{r=0}^{d} \binom{D}{r}. \end{aligned} \tag{11}$$

For inexact solves, a feasible recovered score supplies an upper bound U_t, dual feasible cuts remain global lower bounds, and a global master bound L_t gives

$$0 \le R(f_{\mathrm{inc}}) - \mathrm{OPT} \le U_t - L_t.$$

Proposition C.1(i) proves the certificate and the sufficient accuracy condition $\delta + \beta \le \varepsilon$ for finite termination. Parts (ii) and (iii) of the same proposition give bounded slopes and a data-dependent separation bound for exact search.

Algorithm 1 Exact cutting-plane method for variable selection

Require: Problem data, budget d, tolerance $\varepsilon > 0$, and $z^1 \in \mathcal{Z}_d$.
Ensure: A subset S^*, its optimal test f^*, and a risk interval of width at most ε.

$t \leftarrow 1$, $U \leftarrow +\infty$.
while true **do**
  Solve (8) exactly at z^t; choose μ^t with $0 \le \mu_j^t \le M_j$ as in Proposition C.1(ii).
  if $F(z^t) < U$ **then**
    $(U, z^*) \leftarrow (F(z^t), z^t)$.
  end if
  Retain the cut (9) with index t.
  Solve (10) exactly to obtain (z^{t+1}, η^{t+1}).
  if $U - \eta^{t+1} \le \varepsilon$ **then**
    $S^* \leftarrow \{j : z_j^* = 1\}$.
    Solve (4) for S^*; construct g^* by (17) and set $f^* = g^* \circ \mathrm{Proj}_{S^*}$.
    return $(S^*, f^*, [\eta^{t+1}, U])$.
  end if
  $t \leftarrow t + 1$.
end while

A Proof of the finite representation

Let $x_i \in \mathbb{R}^D$, $i \in [N]$, and let nonempty sets I_0, I_1 partition $[N]$. Let $q_i > 0$, $\sum_{i\in I_k} q_i = 1$, and $\theta_k > 0$. The loss $\ell : \mathbb{R} \to [0,\infty)$ is finite, convex and nondecreasing, with $\ell(0) > 0$; put $\ell_0(s) = \ell(-s)$ and $\ell_1(s) = \ell(s)$. For $S \subseteq [D]$, let Proj_S be coordinate projection; δ_x denotes unit mass at x. Define

$$x_i^S = \mathrm{Proj}_S\, x_i, \qquad c_{ir}^S = \|x_i^S - x_r^S\|_1, \qquad \widehat{P}_k = \sum_{i\in I_k} q_i \delta_{x_i},$$

$$\mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \qquad \mathcal{G}_S = \{g : \mathbb{R}^{|S|} \to \mathbb{R} :\ g \text{ is Borel measurable}\},$$

$$R(f) = \sum_{k=0}^{1} \sup_{P\in\mathcal{U}_k} \mathbb{E}_P \ell_k(f(X)), \qquad V(S) = \inf_{g\in\mathcal{G}_S} R(g \circ \mathrm{Proj}_S).$$

Here W_1 uses ℓ_1 transport cost and finite-first-moment distributions. Set

$$\begin{aligned} T(S) = \inf_{a,u\in\mathbb{R}^N,\ \lambda\in\mathbb{R}_+^2} \quad & J(a,\lambda) := \sum_i q_i a_i + \sum_{k=0}^{1} \theta_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N],\ k = 0,1. \end{aligned} \tag{12}$$

$T(S)$ is the finite value; equality with $V(S)$ is proved below.

Proposition A.1. *For every $S \subseteq [D]$, $V(S) = T(S)$, the infimum defining $T(S)$ is attained, and each optimal finite solution produces a Borel score $g^* : \mathbb{R}^{|S|} \to \mathbb{R}$ with $R(g^* \circ \mathrm{Proj}_S) = V(S)$.*

Proof. Finite transport duality. Fix $u \in \mathbb{R}^N$. Define

$$\mathcal{T}_k(S) = \left\{ \pi \in \mathbb{R}_+^{|I_k|\times N} : \sum_r \pi_{ir} = q_i\ (i \in I_k), \quad \sum_{i\in I_k, r} \pi_{ir} c_{ir}^S \le \theta_k \right\}.$$

The diagonal plan is feasible and $0 \le \pi_{ir} \le q_i$; thus $\mathcal{T}_k(S)$ is nonempty and compact. With $a_i \in \mathbb{R}$, $\lambda_k \ge 0$, and indices $i \in I_k$, $r \in [N]$,

$$\begin{aligned} L_k(\pi, a, \lambda_k; u) &= \sum_{i,r} \pi_{ir} \ell_k(u_r) + \sum_i a_i \left(q_i - \sum_r \pi_{ir} \right) + \lambda_k \left(\theta_k - \sum_{i,r} \pi_{ir} c_{ir}^S \right) \\ &= \sum_i q_i a_i + \theta_k \lambda_k + \sum_{i,r} \pi_{ir} [\ell_k(u_r) - a_i - \lambda_k c_{ir}^S]. \end{aligned} \tag{13}$$

$$\sup_{\pi\ge 0} L_k = \begin{cases} \sum_i q_i a_i + \theta_k \lambda_k, & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S \quad \forall i, r, \\ +\infty, & \text{otherwise.} \end{cases}$$

LP strong duality gives

$$\begin{aligned} \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k, r} \pi_{ir} \ell_k(u_r) = \min_{(a_i)_{i\in I_k},\ \lambda_k \ge 0} \quad & \sum_{i\in I_k} q_i a_i + \theta_k \lambda_k \\ \text{subject to} \quad & \ell_k(u_r) \le a_i + \lambda_k c_{ir}^S, \quad i \in I_k,\ r \in [N]. \end{aligned} \tag{14}$$

$$T(S) = \inf_{u\in\mathbb{R}^N} \sum_{k=0}^{1} \max_{\pi\in\mathcal{T}_k(S)} \sum_{i\in I_k, r} \pi_{ir} \ell_k(u_r).$$

Only the finite LP has been dualized; no minimax exchange is used.

Lower bound: $T(S) \le V(S)$. For $g \in \mathcal{G}_S$ and $\pi \in \mathcal{T}_k(S)$, choose $\tilde{x}_{ir} \in \mathbb{R}^D$ with

$$\mathrm{Proj}_S\, \tilde{x}_{ir} = x_r^S, \qquad \mathrm{Proj}_{S^c}\, \tilde{x}_{ir} = \mathrm{Proj}_{S^c}\, x_i, \qquad \|\tilde{x}_{ir} - x_i\|_1 = c_{ir}^S.$$

$$
\begin{aligned}
P_k^\pi &= \sum_{i\in I_k,r} \pi_{ir}\delta_{\widetilde{x}_{ir}}, \qquad W_1(P_k^\pi,\widehat{P}_k) \le \sum_{i\in I_k,r}\pi_{ir}c_{ir}^S \le \theta_k,\\
\mathbb{E}_{P_k^\pi}\ell_k(g(\mathrm{Proj}_S X)) &= \sum_{i\in I_k,r}\pi_{ir}\ell_k(g(x_r^S)).
\end{aligned}
\tag{15}
$$

$$
\begin{aligned}
V(S) &\ge \inf_{g\in\mathcal{G}_S}\sum_k \max_{\pi\in\mathcal{T}_k(S)}\sum_{i\in I_k,r}\pi_{ir}\ell_k(g(x_r^S))\\
&\ge \inf_{u\in\mathbb{R}^N}\sum_k \max_{\pi\in\mathcal{T}_k(S)}\sum_{i\in I_k,r}\pi_{ir}\ell_k(u_r) = T(S).
\end{aligned}
\tag{16}
$$

Upper bound: $V(S)\le T(S)$. For any finite feasible (a,u,λ) and $y\in\mathbb{R}^{|S|}$, set

$$
\begin{gathered}
E_k(y)=\min_{i\in I_k}\{a_i+\lambda_k\|x_i^S-y\|_1\}, \qquad i_k(y)=\min\arg\min_{i\in I_k}\{a_i+\lambda_k\|x_i^S-y\|_1\},\\
r_k=\|x_{i_k}^S-y\|_1, \qquad \rho=\|x_{i_0}^S-x_{i_1}^S\|_1\le r_0+r_1.\\
g(y)=\begin{cases}(1-\gamma)u_{i_0}+\gamma u_{i_1}, & \gamma=\min\{1,r_0/\rho\}, \quad \rho>0,\\ u_{i_0}, & \rho=0.\end{cases}
\end{gathered}
\tag{17}
$$

For $\rho>0$, $\gamma\rho\le r_0$ and $(1-\gamma)\rho\le r_1$. Convexity and finite feasibility yield

$$
\begin{aligned}
\ell_0(g(y)) &\le (1-\gamma)\ell_0(u_{i_0})+\gamma\ell_0(u_{i_1}) \le a_{i_0}+\lambda_0\gamma\rho \le E_0(y),\\
\ell_1(g(y)) &\le (1-\gamma)\ell_1(u_{i_0})+\gamma\ell_1(u_{i_1}) \le a_{i_1}+\lambda_1(1-\gamma)\rho \le E_1(y).
\end{aligned}
$$

For $\rho=0$, the source–target pairs (i_0,i_0) and (i_1,i_0) give the same two bounds. The finite tie rule makes g Borel, with $|g(y)|\le\max_r|u_r|$; interpolation of the u_r is unnecessary. For $P\in\mathcal{U}_k$, disintegrate an optimal coupling as source-indexed measures ν_i (splitting repeated sources by q_i):

$$
\nu_i(\mathbb{R}^D)=q_i, \qquad \sum_{i\in I_k}\nu_i=P, \qquad \sum_{i\in I_k}\int\|x-x_i\|_1\,\nu_i(dx)\le\theta_k.
$$

$$
\begin{aligned}
\mathbb{E}_P\ell_k(g(\mathrm{Proj}_S X)) &\le \sum_{i\in I_k}\int[a_i+\lambda_k\|\mathrm{Proj}_S\,x-x_i^S\|_1]\,\nu_i(dx)\\
&\le \sum_{i\in I_k} q_i a_i+\theta_k\lambda_k.
\end{aligned}
\tag{18}
$$

$$
R(g\circ\mathrm{Proj}_S)\le J(a,\lambda), \qquad V(S)\le T(S).
$$

The feasible point $(a,u,\lambda)=(\ell(0)\mathbf{1},0,0)$ has value $C=2\ell(0)$. On the closed sublevel $J\le C$,

$$
0\le a_i\le C/q_i, \qquad 0\le\lambda_k\le C/\theta_k.
$$

The loss constraints bound both $\ell(u_r)$ and $\ell(-u_r)$. If ℓ is nonconstant, $\ell(t)\to\infty$ as $t\to\infty$, so u is bounded; if ℓ is constant, restrict to $u=0$. Weierstrass's theorem gives a minimizer, whose extension attains $V(S)$. □

B Proof of the dual reformulation

Let $x_i\in\mathbb{R}^D$, let nonempty I_0,I_1 partition $[N]$, and let $q_i,\theta_k>0$ with $\sum_{i\in I_k}q_i=1$. Let $\ell:\mathbb{R}\to[0,\infty)$ be finite, convex and nondecreasing, with $\ell(0)>0$; write $\ell_0(s)=\ell(-s)$ and $\ell_1(s)=\ell(s)$. Set

$$
\Delta_{ir,j}=|x_{ij}-x_{rj}|, \qquad c_{ir}^S=\sum_{j\in S}\Delta_{ir,j}, \qquad J(a,\lambda)=\sum_i q_i a_i+\sum_k\theta_k\lambda_k.
$$

Let $V(S)$ and $F(z)$ be the values of (4) and (5), respectively. For $\Pi\in\mathbb{R}_+^{N\times N}$, define

$$H(\Pi)=\sum_{r=1}^{N}\inf_{s\in\mathbb{R}}\sum_{k=0}^{1}\sum_{i\in I_k}\Pi_{ir}\ell_k(s),\qquad d_{kj}(\Pi)=\sum_{i\in I_k,r}\Pi_{ir}\Delta_{ir,j}.$$

$$\mathcal{D}=\left\{(\Pi,\alpha,\mu)\ge 0:\ \begin{array}{ll}\sum_r \Pi_{ir}=\alpha q_i & (i\in[N]),\\ \sum_j (d_{kj}(\Pi)-\theta_k\mu_j)_+\le \alpha\theta_k & (k=0,1)\end{array}\right\},\qquad (v)_+=\max\{v,0\}. \tag{19}$$

Proposition B.1 (Exact extension and dual representation)**.** (i) *For* $z=\mathbf{1}_S$, (4) *and* (5) *have the same feasible* (a,u,λ) *after eliminating* w*, with the same objective. Hence* $F(\mathbf{1}_S)=V(S)$.

(ii) *For every* $z\in[0,1]^D$*, the primal and dual optima are attained and*

$$0<F(z)=\max_{\substack{(\Pi,\alpha,\mu)\in\mathcal{D}\\ \alpha+\mu^\top z=1}} H(\Pi)\le 2\ell(0). \tag{20}$$

(iii) *The equivalent affine dual is*

$$F(z)=\max_{(\Pi,\alpha,\mu)\in\mathcal{D}}\{2\sqrt{H(\Pi)}-\alpha-\mu^\top z\}. \tag{21}$$

A normalized optimizer $(\bar\Pi,\bar\alpha,\bar\mu)$ *yields an affine optimizer*

$$(\Pi,\alpha,\mu)=F(z)(\bar\Pi,\bar\alpha,\bar\mu),\qquad H(\Pi)=F(z)^2. \tag{22}$$

The function F *is convex. Every affine optimizer* (Π^t,α^t,μ^t) *at* z^t *satisfies*

$$F(z)\ge F(z^t)-(\mu^t)^\top(z-z^t)\quad(z\in[0,1]^D),\qquad -\mu^t\in\partial F(z^t).$$

Proof. (i) Fix $z=\mathbf{1}_S$. For a feasible point of (4), $a_i\ge\ell_k(u_i)\ge 0$. With $w_k=\lambda_k z$,

$$\ell_k(u_r)\le a_i+\sum_j\Delta_{ir,j}w_{kj},\qquad \sum_k\theta_k w_{kj}=z_j\sum_k\theta_k\lambda_k\le z_jJ(a,\lambda).$$

This gives a feasible point of (5) with unchanged objective. Conversely, feasibility of (5) implies

$$w_{kj}=0\quad(j\notin S),\qquad 0\le w_{kj}\le\lambda_k\quad(j\in S),$$
$$\ell_k(u_r)\le a_i+\sum_{j\in S}\Delta_{ir,j}w_{kj}\le a_i+\lambda_k c^S_{ir}.$$

Thus (a,u,λ) is feasible for (4) with unchanged J. The two maps prove (i).

(ii) Assign nonnegative multipliers $\Pi_{ir},\beta_{kj},\mu_j$ to the loss, price-cap and selector inequalities in (5). Keeping $\lambda,w\ge 0$ and writing $\alpha=1-\mu^\top z$, the Lagrangian is

$$\begin{aligned}\mathcal{L}&=J+\sum_{k,i\in I_k,r}\Pi_{ir}\left[\ell_k(u_r)-a_i-\sum_j\Delta_{ir,j}w_{kj}\right]+\sum_{k,j}\beta_{kj}(w_{kj}-\lambda_k)\\&\quad+\sum_j\mu_j\left[\sum_k\theta_k w_{kj}-z_jJ\right]\\&=\sum_i a_i\left(\alpha q_i-\sum_r\Pi_{ir}\right)+\sum_{k,i\in I_k,r}\Pi_{ir}\ell_k(u_r)\\&\quad+\sum_k\lambda_k\left(\alpha\theta_k-\sum_j\beta_{kj}\right)+\sum_{k,j}w_{kj}[\beta_{kj}+\theta_k\mu_j-d_{kj}(\Pi)].\end{aligned} \tag{23}$$

The infimum over $a, u \in \mathbb{R}^N$ and $\lambda, w \geq 0$ equals $H(\Pi)$ exactly under the conditions

$$\sum_r \Pi_{ir} = \alpha q_i, \qquad \sum_j \beta_{kj} \leq \alpha\theta_k, \qquad \beta_{kj} \geq (d_{kj}(\Pi) - \theta_k\mu_j)_+. \tag{24}$$

The row sums give $\alpha \geq 0$; eliminating β yields $\mathcal{D}$ and $\alpha + \mu^\top z = 1$. For $J_0 = \{j : z_j = 0\}$, the primal forces $w_{kj} = 0$. After deleting these variables and their selector rows, the point

$$u = 0, \qquad a_i = \ell(0) + 1, \qquad \lambda_k = 1, \qquad w_{kj} = z_j/2 \quad (j \notin J_0).$$

All remaining inequalities are strict, including

$$\sum_k \theta_k w_{kj} = \frac{z_j}{2} \sum_k \theta_k < z_j \left[2(\ell(0) + 1) + \sum_k \theta_k \right] = z_j J.$$

Slater's theorem gives strong duality and dual attainment for the reduced problem, including $J_0 = [D]$. Restoring

$$\beta_{kj} = 0, \qquad \mu_j = \max_{k=0,1} d_{kj}(\Pi)/\theta_k \quad (j \in J_0)$$

preserves the objective and normalization and satisfies (24). The primal sublevel $J \leq C := 2\ell(0)$ is nonempty. Since $w_k \leq \lambda_k \mathbf{1}$, its (a, u, λ) satisfy the bounds in the proof of Proposition A.1 with $S = [D]$; w is bounded as well. The same compactness argument gives primal attainment and $F(z) \leq C$. If $F(z) = 0$, nonnegativity gives $a = \lambda = w = 0$, contradicting

$$0 < 2\ell(0) \leq \ell(-u_r) + \ell(u_r) \leq 0.$$

(iii) The set $\mathcal{D}$ is a cone and $H(\tau\Pi) = \tau H(\Pi) \geq 0$ for $\tau \geq 0$. For $(\Pi, \alpha, \mu) \in \mathcal{D}$, put $b = \alpha + \mu^\top z$. If $b > 0$, (ii) gives $H(\Pi)/b \leq F(z)$ and

$$2\sqrt{H(\Pi)} - b = \frac{H(\Pi)}{b} - \left(\sqrt{\frac{H(\Pi)}{b}} - \sqrt{b} \right)^2 \leq F(z). \tag{25}$$

If $b = 0$, then $\alpha = 0$, $\Pi = 0$, and the affine objective is zero. Conversely, for a normalized optimizer,

$$H(F(z)\bar{\Pi}) = F(z)^2, \qquad F(z)(\bar{\alpha} + \bar{\mu}^\top z) = F(z), \qquad 2\sqrt{H(F(z)\bar{\Pi})} - F(z) = F(z).$$

This proves (21) and (22). As a supremum of affine functions, F is convex, and an optimizer at z^t gives

$$F(z) \geq 2\sqrt{H(\Pi^t)} - \alpha^t - (\mu^t)^\top z = F(z^t) - (\mu^t)^\top (z - z^t).$$

The rescaling gives $-\mu^t = -F(z^t)\bar{\mu}^t$. Since H is a nonnegative infimum of linear functions, both H and $\sqrt{H}$ are concave; the fixed-z dual is a convex program. □

C Proof of the search guarantees

For the value function F and dual objects $H, \mathcal{D}$ of Proposition B.1, set

$$\mathcal{Z}_d = \{\mathbf{1}_S : S \subseteq [D],\ |S| \leq d\}, \qquad \mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z).$$

At $z^s = \mathbf{1}_{S_s} \in \mathcal{Z}_d$, let p_s be a feasible objective of (5), with recovered score $f_s = g_s \circ \mathrm{Proj}_{S_s}$. For $(\Pi^s, \alpha_s, \mu^s) \in \mathcal{D}$ and a certified bound $0 \leq h_s \leq H(\Pi^s)$, define

$$c_s(z) = 2\sqrt{h_s} - \alpha_s - (\mu^s)^\top z, \qquad m_t(z) = \max\{0, c_1(z), \dots, c_t(z)\},$$
$$U_t = \min_{s \le t} p_s, \qquad s_t \in \arg\min_{s \le t} p_s, \qquad f_{\text{inc}} = f_{s_t}.$$

Let L_t be a global lower bound on $\min_{z \in \mathcal{Z}_d} m_t(z)$ and let (z^{t+1}, η^{t+1}) be a feasible master point: $z^{t+1} \in \mathcal{Z}_d$, $\eta^{t+1} \ge m_t(z^{t+1})$.

Proposition C.1 (Risk certificate and finite search)**.** (i) *For every t,*

$$L_t \le \text{OPT} \le R(f_{\text{inc}}) \le U_t, \qquad 0 \le R(f_{\text{inc}}) - \text{OPT} \le U_t - L_t. \tag{26}$$

If, at every iteration, for $\delta, \beta \ge 0$ and $\varepsilon > 0$,

$$p_s - c_s(z^s) \le \delta \quad (s \le t), \qquad \eta^{t+1} - L_t \le \beta, \qquad \delta + \beta \le \varepsilon,$$

then stopping at $U_t - L_t \le \varepsilon$ requires at most $|\mathcal{Z}_d|$ evaluations.
(ii) *Put*

$$C = 2\ell(0), \qquad A_{kj} = \sum_{i \in I_k} q_i \max_{r \in [N]} \Delta_{ir,j}, \qquad M_j = C \max_{k=0,1} \frac{A_{kj}}{\theta_k}. \tag{27}$$

At every $z \in [0,1]^D$ there exists an affine optimizer with $0 \le \mu_j \le M_j$. Consequently,

$$|F(z) - F(z')| \le \sum_j M_j |z_j - z'_j| \qquad (z, z' \in [0,1]^D).$$

(iii) *Under exact solves and the bounded optimal multipliers in* (ii)*, Algorithm 1 returns f^* at iteration T with*

$$0 \le R(f^*) - \text{OPT} \le U_T - \eta^{T+1} \le \varepsilon. \tag{28}$$

Any two actually evaluated selections satisfy

$$\sum_{j \in S_t \setminus S_s} M_j > \varepsilon \qquad (s < t). \tag{29}$$

For the pseudodistance $d_M(S, S') = \sum_{j \in S \triangle S'} M_j$, define

$$\mathcal{P}_\varepsilon = \max\left\{ |\mathcal{A}| : \begin{array}{l} \mathcal{A} \subseteq \{S \subseteq [D] : |S| \le d\}, \\ d_M(S, S') > \varepsilon \quad (S \ne S',\ S, S' \in \mathcal{A}) \end{array} \right\}.$$

Then, allowing $M_j = 0$,

$$T \le \mathcal{P}_\varepsilon \le \sum_{r=0}^{d} \binom{D}{r}, \qquad N_{\text{dual}} = N_{\text{master}} = T, \qquad N_{\text{recovery}} = 1. \tag{30}$$

Here N_{recovery} counts the final fixed-selection primal solve and score construction together.

Proof. (i) Proposition B.1(i) and the extension in Proposition A.1 give $R(f_s) \le p_s$. Dual feasibility and $h_s \le H(\Pi^s)$ give $0 \le m_t \le F$, hence

$$L_t \le \min_{z \in \mathcal{Z}_d} m_t(z) \le \text{OPT} \le R(f_{s_t}) \le p_{s_t} = U_t.$$

If $z^{t+1} = z^s$ for some $s \le t$, then

$$U_t - L_t \le p_s - c_s(z^s) + \eta^{t+1} - L_t \le \delta + \beta \le \varepsilon.$$

Thus every nonterminal iteration selects a previously unevaluated point, proving the evaluation bound.

(ii) Scale a normalized optimizer as in (22). Then $0 \le \alpha = F(z)\bar{\alpha} \le C$ and

$$d_{kj}(\Pi) = \sum_{i \in I_k, r} \Pi_{ir} \Delta_{ir,j} \le \sum_{i \in I_k} \alpha q_i \max_r \Delta_{ir,j} = \alpha A_{kj}.$$

Set $b_j = \max_k d_{kj}(\Pi)/\theta_k$ and $\mu'_j = \min\{\mu_j, b_j\}$. If $\mu_j > b_j$, then

$$(d_{kj}(\Pi) - \theta_k \mu'_j)_+ = (d_{kj}(\Pi) - \theta_k \mu_j)_+ = 0 \quad (k = 0, 1).$$

All other coordinates are unchanged, so $(\Pi, \alpha, \mu') \in \mathcal{D}$ and

$$F(z) \ge 2\sqrt{H(\Pi)} - \alpha - (\mu')^\top z \ge 2\sqrt{H(\Pi)} - \alpha - \mu^\top z = F(z),$$
$$0 \le \mu'_j \le b_j \le \alpha \max_k A_{kj}/\theta_k \le M_j.$$

Applying the resulting supporting cuts at z and z' gives

$$-\sum_j M_j |z_j - z'_j| \le F(z) - F(z') \le \sum_j M_j |z_j - z'_j|.$$

(iii) In (i), take $p_s = c_s(z^s) = F(z^s)$, $\delta = \beta = 0$, and $L_t = \eta^{t+1}$. The recovered optimum has risk U_T, proving (28). For $s < t$, the decision to evaluate z^t requires

$$c_s(z^t) \le \eta^t < U_{t-1} - \varepsilon \le F(z^s) - \varepsilon,$$
$$\varepsilon < (\mu^s)^\top (z^t - z^s) = \sum_{j \in S_t \setminus S_s} \mu_j^s - \sum_{j \in S_s \setminus S_t} \mu_j^s \le \sum_{j \in S_t \setminus S_s} M_j \le d_M(S_s, S_t).$$

The evaluated selections therefore form an ε-packing, yielding $T \le \mathcal{P}_\varepsilon \le |\mathcal{Z}_d|$. Each iteration solves one dual and one master; the last step solves one fixed-selection primal and constructs its score, proving (30). □